

Hyperspectral Image Classification With Contrastive Self-Supervised Learning Under Limited Labeled Samples

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Abstract—Hyperspectral image (HSI) classification is an active research topic in remote sensing. Supervised learning-based methods have been widely used in HSI classification tasks due to their powerful feature extraction capabilities for cases of sufficiently labeled samples. However, practical applications often have limited samples with accurate labels due to the high cost of labeling or unreliable visual interpretation. We introduce a contrastive self-supervised learning (SSL) algorithm to achieve HSI classification for problems with few labeled samples. First, a new HSI-specific augmentation module is developed to generate sample pairs. Then, a contrastive SSL model based on Siamese networks is used to extract features from these easily accessible sample pairs. Finally, the labeled samples are taken to fine-tune the parameters of the classification model to boost classification performance. Tests of the contrastive self-supervised algorithm have been performed on two widely used HSI datasets. The experimental results reveal that the proposed algorithm requires a few labeled samples to achieve superior performance.

Index Terms—Contrastive learning, hyperspectral image (HSI) classification, limited labeled samples, self-supervised learning (SSL).

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain spatial and spectral information, which has gained increased applications in fields such as geology, vegetation ecology, agriculture, and oceans over the past decade [1]. Object classification is the core task of these applications; thus, many classification methods using different deep learning techniques have been proposed for HSI classification.

The prerequisite for achieving accurate HSI classification is to obtain effective features. Deep convolutional neural networks (CNNs) based on supervised learning yield superior performance in extracting spatial and spectral features from HSIs and dominate the field of HSI classification [2].

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Kang *et al.* [3] proposed a dual-path network structure combining a residual network and CNN. Zhong *et al.* [4] proposed a spectral space residual network by fusing the ideas of residual neural networks and 3-D CNNs. Ghaderizadeh *et al.* [5] presented a hybrid CNN model of 3-D CNN and 2-D CNN for HSI classification. The above methods based on supervised learning extract high-quality spatial and spectral features. However, the methods of supervised learning for image feature extraction require numerous labeled samples for training. The process of field measurement and manual labeling of an HSI is time-consuming and error-prone. As a result, the image representation learning methods, which require few labeled samples or even do not need them, are highly desired in remote sensing. There are some new attempts based on transfer learning for the HSI classification task. Liu *et al.* [6] proposed a domain adaptation method called statistical fusion. Qu *et al.* [7] proposed a transfer learning algorithm that bridges the gap by projecting the source and target domains into a shared abundance space between different domains. In addition, some heterogeneous transfer learning-based algorithms such as HT-CNN [8] and HFSL [9] also have remarkable classification performance.

Currently, a popular training paradigm is self-supervised learning (SSL) without labels [10]. Among them, contrastive learning has recently achieved state-of-the-art performance. The core idea of contrastive learning is that different augmented views of the same image are brought together, and the augmented views from different images are pushed apart. Contrastive learning methods, such as SimCLR [11], MOCO [12], and BOYL [13], perform well on the ImageNet classification task. However, these methods are usually developed for color images to learn spatial image representation, and they cannot fit the HSIs to effectively extract spectral-spatial representation due to the HSI characteristic of high spectral dimensionality (far greater than three channels of color images).

An HSI-specific contrastive learning approach is proposed for few labeled samples classification. Not only does the method have superior classification performance, but it also requires only a few labeled samples (see Fig. 1). The proposed hyperspectral classification method consists of three parts, as presented in Fig. 2. First, an augmentation module is specially designed for HSIs with high spectral dimensions. Image augmentation is equally important for SSL compared with supervised learning. The image augmentation module

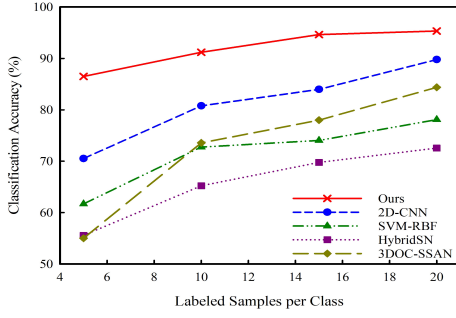


Fig. 1. Testing the classification accuracy of the various methods on the University of Pavia dataset.

generates multiple views of the same HSI and increases the number of samples. Second, the SSL model framework with a Siamese network structure [14] extracts the features of HSIs with numerous unlabeled training samples. The Siamese network model based on contrast learning can yield effective spatial and spectral representations using contrastive loss, which pulls together similar HSI patch pairs while pushing apart dissimilar pairs. The third part is a downstream classification model framework. We transferred the pre-trained model parameters and fine-tuned them through a downstream supervised learning classification task with a few labeled samples. The main contributions of this work are listed as follows.

- 1) A contrastive learning approach is introduced for hyperspectral classification, which can extract robust spatial-spectral features from readily available unlabeled samples.
- 2) We developed an integrated image augmentation module for multiband HSIs in the field of contrastive learning. This module improves the robustness and overall performance of the HSI classification model while increasing the number of samples.

II. METHODOLOGY

A. Hyperspectral Image Augmentation Module

Image augmentation plays a crucial role in contrastive learning. However, systematic image augmentation methods are lacking in HSI classification based on contrastive learning, which drastically limits the application of contrastive learning methods in the hyperspectral field. Therefore, as presented in Fig. 3, we developed a family of HSI-specific data augmentation and composed them stochastically.

Fig. 3(a) illustrates the combined operation of random cropping and resizing. Specifically, given an HSI patch $I \in \mathbb{R}^{W \times H \times C}$, the cropped HSI patch I_c can be obtained as follows:

$$I_c = I[x_0 : x_0 + \lfloor rW \rfloor, y_0 : y_0 + \lfloor rH \rfloor, 1 : C] \quad (1)$$

where the operator $\lfloor \cdot \rfloor$ returns a slice of I specified by a range of indices, and the operator $\lfloor \cdot \rfloor$ represents round toward negative infinity. r is the scale ratio of the cropped patch to the original patch I , which is less than 1. x_0 is the random integer range from 1 to $W - \lfloor rW \rfloor$, and y_0 is the random integer range from 1 to $H - \lfloor rH \rfloor$. The cropped patch I_c is

then resized to obtain the transformed image $\tilde{I}$ with the same size as the original patch I . Usually, the policy of applying a transformation with a certain probability is employed to increase the diversity of the augmented data. Random cropping is performed with p_1 probability, given by the following:

$$\tilde{I}_1 = \begin{cases} \text{resize}(I_c), & \text{if } \mu < p_1 \\ I, & \text{otherwise} \end{cases} \quad (2)$$

where $\text{resize}(\cdot)$ represents the operator that resizes the cropped patch to the size of the original patch I and μ is sampled from the uniform distribution (0, 1).

The horizontal flipping operator of the HSI patch is displayed in Fig. 3(b). Random flipping follows random cropping (with resizing) with p_2 probability, given by the following:

$$\tilde{I}_2 = \begin{cases} \text{h_flip}(\tilde{I}_1), & \text{if } \mu < p_2 \\ \tilde{I}_1, & \text{otherwise} \end{cases} \quad (3)$$

where $\text{h_flip}(\cdot)$ represents the operator that flips the HSI patch in the left-right direction.

Fig. 3(c) illustrates the color jitter operation after the flipping operator. The pixel value of each band of the HSI patch is jointly adjusted by the brightness and contrast factors as follows:

$$\tilde{I}_3^c(i, j) = \begin{cases} \alpha \times \tilde{I}_2^c(i, j) + \beta, & \text{if } \mu < p_3 \\ \tilde{I}_2^c(i, j), & \text{otherwise} \end{cases} \quad (4)$$

where $\tilde{I}_2^c(i, j)$ is the spectral intensity corresponding to the spatial location (i, j) at the c th band of HSI patch $\tilde{I}_2$, α is the contrast factor, and β is the brightness factor. Finally, p_3 is the probability that the color jitter is applied to the HSI augmentation.

The last Gaussian blur operation is illustrated in Fig. 3(d). An $r \times r$ Gaussian convolutional kernel acts on the transformed patch $\tilde{I}_3$. When the random number μ is less than p_4 , it is formulated as follows:

$$\tilde{I}_4^c(i, j) = \sum_{k=1}^r \sum_{h=1}^r \left(\sigma(h, k) \times \tilde{I}_3^c \left(i + h - \frac{r+1}{2}, j + k - \frac{r+1}{2} \right) \right) \quad (5)$$

where $\sigma(h, k)$ denotes the weight value of the Gaussian convolutional kernel and r is the Gaussian convolutional kernel size.

The process is repeated twice. A minibatch of N samples $\{I_1, I_2, \dots, I_n\}$ is augmented with $2N$ samples $\{I_1^1, I_1^2, \dots, I_n^1, I_n^2\}$. In addition, (I_i^1, I_i^2) forms a positive sample pair, and the remaining $2(N-1)$ augmented samples form $2(N-1)$ negative sample pairs $(I_i^1, I_n^k)_{[i \neq n, k \in \{1, 2\}]}$.

B. Learning Representation Using Contrastive Learning

The pretext task is to maximize instance similarity, and the core of this method is to compare the similarity between numerous augmented unlabeled images that motivate the model to learn more advanced representations. First, the ResNet-18-based [15] encoder is applied to extract the spatial and spectral features from the augmented samples of the minibatch. Moreover, (z_i^1, z_i^2) is the output vector of the

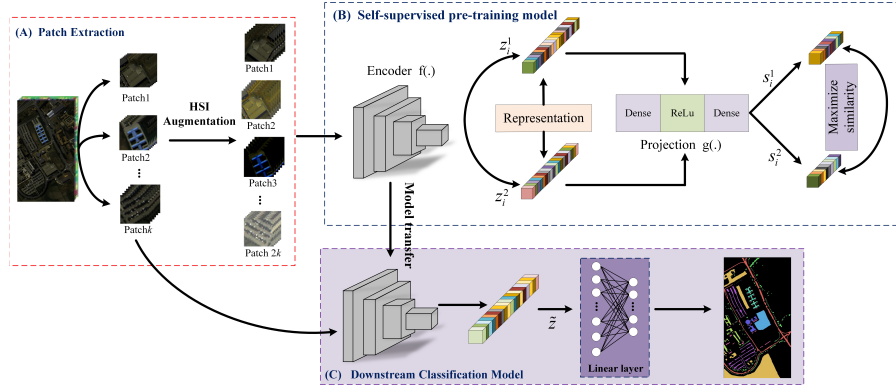


Fig. 2. Framework of the contrastive SSL method for HSI classification, where part A is the data augmentation module for acquiring different views, part B is the model pre-training procedure, and part C is the model fine-tuning and testing procedure.

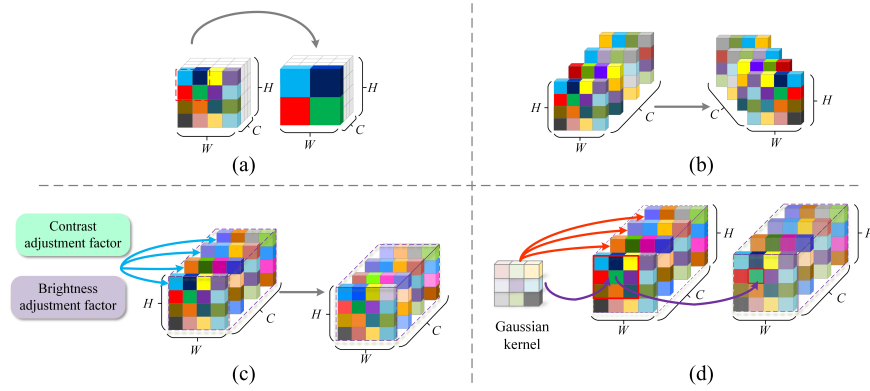


Fig. 3. Four kinds of HSI transformation operations. (a) Crop and Resize. (b) Horizontal Flip. (c) Color Jitter. (d) Gaussian Blur.

TABLE I
NETWORK STRUCTURE OF THE PRE-TRAINED MODEL

Conv1	Conv2		Conv3		Conv4		Conv5		AVGpool	Dense1	Dense2
$[3 \times 3, 64]$	$3 \times 3, 64$	$\times 2$	$3 \times 3, 128$	$\times 2$	$3 \times 3, 256$	$\times 2$	$3 \times 3, 512$	$\times 2$	AdaptiveAvgPool	(256)	(128)

positive sample pair (I_i^1, I_i^2) after applying the ResNet18-based encoder. Then, the projection head $g(\cdot)$ containing two dense layers and a rectified linear unit (ReLU) activation function is used to form and maintain more information from (z_i^1, z_i^2) that might be useful for downstream tasks. In addition, $S_i = g(z_i) = \omega^{(2)}\delta(\omega^{(1)}z_i)$ is the output result of the projection head, where δ is the ReLU activation function. The cosine similarity is taken to measure the related similarity between patches I_i^1 and I_i^2 , formulated as follows:

$$\text{similarity}(I_i^1, I_i^2) = \cos(S_i^1, S_i^2) = (S_i^1)^T S_i^2 / \|S_i^1\| \|S_i^2\| \quad (6)$$

where $\|\cdot\|$ represents the norm of S . Thus, when I_i^1 and I_i^2 are more similar, the value of the cosine similarity is greater. The contrastive loss, NT-Xent loss [16], is generated as follows:

$$l_{I_i^1, I_i^2} = -\log \frac{\exp(\cos(S_i^1, S_i^2)) / \tau}{\sum_{n=1}^N \mathbb{I}_{[n \neq i \& k \neq 1]} \exp(\cos(S_i^1, S_n^k) / \tau)} \quad (7)$$

where τ denotes a temperature parameter set to 0.5 in the experiment, $k \in \{1, 2\}$, and $\mathbb{I}_{[n \neq i \& k \neq 1]} \in \{0, 1\}$ is an indicator

function. When $n \neq i$ and $k \neq 1$, the evaluation is 1. The model parameters can be updated by maximizing the similarity of positive sample pairs and minimizing the similarity of negative sample pairs. Both (I_i^1, I_i^2) and (I_i^2, I_i^1) are accounted for in the total loss of the minibatch using the following equation:

$$L = \frac{1}{2N} \sum_{i=1}^N [l_{I_i^1, I_i^2} + l_{I_i^2, I_i^1}]. \quad (8)$$

Table I shows the details of the network structure of the pre-trained model.

C. Fine-Tuning Using Limited Labeled Samples

In this section, the pre-trained ResNet18-based encoder is transferred to the downstream supervised learning process. We used limited labeled samples to fine-tune the pre-trained ResNet18-based encoder and train a new classifier. Specifically, in the retraining step, limited labeled samples are

TABLE II
CLASSIFICATION ACCURACY (%) UNDER DIFFERENT
AUGMENTATION POLICY

Augmentation Method	OA	AA	Kappa
Crop	88.78	89.6	86.01
Crop+Flip	86.95	89.3	84.55
Crop+Blur	89.43	89.16	86.25
Crop+Color jitter	91.83	91.02	89.3
Crop+Flip+Blur	83.27	85.9	78.6
Crop+Flip+Color jitter	89.18	89.01	85.96
Crop+Blur+Color jitter	90.28	89.21	88.3

randomly selected as training datasets for supervised learning to fine-tune the previous pre-trained encoder parameters and train a new lightweight fully connected layer to achieve a classifier function. For insufficient labeled samples, fine-tuning the pre-trained encoder can effectively overcome overfitting and achieve higher classification performance.

III. EXPERIMENT AND DISCUSSION

A. Hyperspectral Image Dataset

The classification performance is evaluated in the experiments on two publicly available hyperspectral datasets: the University of Pavia (UP) and Houston 2013 (HS) datasets.

The UP scene dataset is a part of hyperspectral data taken by ROSIS of Germany on the city of Pavia in Italy. The image size is $610 \times 340 \times 103$ with nine classes of interest, and the spectral coverage ranges from 0.43 to 0.86 μm .

The HS dataset is an image that contains information about the University of Houston campus and its surrounding neighborhoods, collected by the ITRES CASI-1500 sensor. The image has 144 spectral bands of size 349×1905 with 15 ground-truth classes.

We compared our method with several state-of-the-art methods. We selected 80% of the samples for each dataset as the unlabeled training dataset for contrastive SSL pretext tasks. In downstream supervised learning, the number of training samples is 1% of the total samples in experiment I. We compared our method with the most widely used supervised learning methods, SVM-RBF, 2-D CNN [17], HybridSN [18], and 3DOC-SSAN [2]. To evaluate the performance under limited labeled samples, in experiment II, five labeled samples from each class are randomly selected as the training samples for fine-tuning, and we compared it with an SSL method called DMVL [19].

B. Experimental Setup

The complete process consists of two parts. In the feature extraction part of contrastive SSL, p_1 and p_3 in the image augmentation module are set to 1 and 0.8, respectively. We set the epoch to 500, the patch size to 21×21 , and the batch size of the model to 256. The initial learning rate is set to 0.001 and reduced it by half after epoch 300. The Adam optimizer is used to update the network parameters of the model [20]. In fine-tuning the supervised learning, different batch sizes are set according to the number of labeled samples.

For experiment I, both the UP and HS datasets are set to 64 for the batch size. For experiment II, the batch size is set to 10. The Adam optimizer is also used to optimize the SoftMax cross-entropy target. We set the learning rate to 0.001 for the classifier and a small one ($\text{LR} \times 0.1$) for the encoder.

We introduced overall accuracy (OA), average accuracy (AA), and Kappa to evaluate the classification performance. The OA represents the number of correctly classified samples in all test samples, and the AA represents the average of the sum of the accuracy for each category. Finally, Kappa represents the strong consistent mutual information between the ground truth and classification maps.

C. Data Augmentation Details

The combination of different data augmentations significantly contributes to the model learning good representations. Therefore, we study the classification performance of the proposed method when combining different augmentations. Half of the samples are selected for model pre-training, and the rest are used as the test dataset. In the fine-tuning procedure, ten labeled samples from each class are randomly selected as the training samples for the supervised classifier. Table II shows classification performance using a different data augmentation policy on the UP dataset. It is obvious that random cropping is a simple and effective augmentation operation for contrastive learning. The model yields optimal classification performance when random cropping combines with color jitter. Each HSI patch I is sequentially applied to the above random transformations. In addition, we have further explored the parameters within the data augmentation. There is a weak correlation between the cropping ratio and the classification accuracy, and a better classification accuracy can be obtained when the cropping ratio is set to 0.95. Contrast and brightness are sampled in smaller intervals to ensure that the model extracts more robust features. The model has the best classification accuracy when the sampling intervals of contrast and brightness are set to (0.8–1.2) and (0–500) (details are shown in the Appendix). Therefore, a combination of random cropping and color jitter augmentation is applied to the pre-training procedure of the model, the cropping ratio is set to 0.95, and the contrast and brightness are randomly sampled from the interval (0.8, 1.2) and (0, 500), respectively.

D. Analysis of Experimental Result

Table III lists the OA, AA, and Kappa score of the SVM-RBF, 2-D CNN, HybridSN, 3DOC-SSAN, DMVL, and the proposed method for different numbers of labeled samples in the UP and HS datasets. The following analysis is made based on Table III.

In Table III, the proposed method has a better classification performance compared to other methods. In experiment I, 1% labeled samples are used for fine-tuning, and the OA of the proposed method (97.69%) on the UP dataset is 2% higher than 2-D CNN (95.67%), 5.51% higher than 3DOC-SSAN (92.18%), over 10% higher than SVM-RBF (87.63%), and nearly 8% higher than HybridSN (89.88%). The proposed method has a significant improvement on the

TABLE III
TESTING ACCURACY OF SVM-RBF, 2-D CNN, HYBRIDSN, 3DOC-SSAN, DMVL, AND PROPOSED METHOD WITH DIFFERENT NUMBERS OF TRAINING SAMPLES ON UP AND HS

Dataset	Indicator	Experiment I					Experiment II		
	Training samples	Labeled Fraction 1%					N = 5		
	Class	SVM-RBF	2D-CNN	HybridSN	3DOC-SSAN	Proposed method	DMVL+SVM	DMVL+RF	Proposed method
Pavia University	OA (%)	87.63±1.01	95.67±1.8	89.88±0.8	92.18±2.23	97.69±0.73	86.96	85.70	87.91
	AA (%)	82.87±1.99	92.13±2.1	84.53±1.94	86.33±2.01	96.21±1.02	80.10	81.22	83.57
	Kappa (%)	93.4±1.83	94.3±2.42	86.4±1.17	89.73±1.84	96.95±0.75	82.46	81.07	84.13
Houston	OA (%)	75.84±0.11	81.05±1.12	72.81±2.68	78.63±2.29	88.25±1.8	78.55	76.65	79.32
	AA (%)	75.23±1.78	79.69±2.97	68.57±1.99	76.49±2.23	88.74±1.54	80.44	78.95	81.35
	Kappa (%)	73.82±1.78	79.5±2.31	70.54±1.76	76.88±2.44	87.3±1.95	76.81	74.75	77.67

Note: N represents the number of labeled samples per class

HS dataset. The OA is 88.25%, which is 15.44% higher than HybridSN (72.81%), 12% higher than SVM-RBF (75.84%), 9.62% higher than 3DOC-SSAN (78.63%), and 7.2% higher than 2-D CNN (81.05%). In experiment II, only five labeled samples per class are selected as the training samples, and the OA on the UP dataset is 87.91%, which is improved by 2.21% compared to the DMVL + RF (85.7%) and 0.95% compared to DMVL + SVM (86.96%). On the HS dataset, the OA of the proposed method (79.32%) is 2.67% higher than DMVL + RF (76.65%) and 0.95% higher than DMVL + SVM (78.55%). However, it can be observed that the classification performance of HybridSN is not so satisfactory and even has the worst classification accuracy on the HS dataset. It indicates that a complex network model trained with a few samples will not have a superior classification performance, instead of overfitting.

IV. CONCLUSION

In this letter, a new data augmentation module is developed for HSIs. Besides, a contractive SSL method is applied to extract the spatial and spectral features of HSIs, and limited labeled samples are used to fine-tune the pre-trained model to classify HSIs. The classification results on the UP and HS datasets indicate that even with a few labeled samples, the classification task can be completed well. Under the condition of limited labeled samples, contractive SSL exhibits remarkable potential for the HSI classification task compared to the supervised learning methods. Future work should focus on studying new SSL methods developed in hyperspectral classification to facilitate hyperspectral classification for applications in geology, vegetation ecology, agriculture, and oceans.

REFERENCES

- [1] J. Li *et al.*, "Multiple feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1592–1606, Mar. 2015.
- [2] X. Tang *et al.*, "Hyperspectral image classification based on 3-D octave convolution with spatial-spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.
- [3] X. Kang, B. Zhuo, and P. Duan, "Dual-path network-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 447–451, Mar. 2019.
- [4] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
- [5] S. Ghaderizadeh, D. Abbasi-Moghadam, A. Sharifi, N. Zhao, and A. Tariq, "Hyperspectral image classification using a hybrid 3D-2D convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7570–7588, 2021.
- [6] X. Liu, S. Jia, M. Xu, and J. Zhu, "Statistical fusion-based transfer learning for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 1108–1111.
- [7] Y. Qu, R. K. Baghbaderani, W. Li, L. Gao, Y. Zhang, and H. Qi, "Physically constrained transfer learning through shared abundance space for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10455–10472, Dec. 2021, doi: 10.1109/TGRS.2020.3045790.
- [8] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.
- [9] Y. Wang *et al.*, "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [10] I. Misra and L. van der Maaten, "Self-supervised learning of pretext-invariant representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6707–6717.
- [11] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
- [12] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.
- [13] J.-B. Grill *et al.*, "Bootstrap your own latent: A new approach to self-supervised learning," 2020, *arXiv:2006.07733*.
- [14] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *Proc. Deep Learn. Workshop (ICML)*, Lille, France, 2015, pp. 1–8.
- [15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [16] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.
- [17] L. Zhao, J. Yi, X. Li, W. Hu, J. Wu, and G. Zhang, "Compact band weighting module based on attention-driven for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9540–9552, Nov. 2021.
- [18] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3D–2D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
- [19] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.
- [20] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.